AERO50009 Propulsion and Turbomachinery
Tutorial Sheet 3

In this tutorial you are required to design a compressor stage giving the highest possible stage pressure ratio p_{03}/p_{01} under a set of given constraints. The deviation is negligible: the inlet and outlet flow angles equal the blade angles. The stage is repeating (check the lecture notes if unclear): $\alpha_1 = \alpha_3$. The following parameters are prescribed and cannot be changed: flow coefficient $\phi = 0.6$ (it is defined as $\phi = V_{\text{ax}}/U$), specific heat ratio $\gamma = 1.4$, work done factor $\lambda = 0.9$, efficiency $\eta = 0.85$ and the blade Mach number $U/c_0 = 0.4$.

The blade outlet angles $\alpha_1 = \alpha_3$ and α_2' can be varied to achieve the highest possible p_{03}/p_{01}, however the de Haller numbers for the rotor and the stator, V_2'/V_1' and V_3/V_2, should not be below 0.72.

Open the Axial compressor stage tool from the Web tools page (also linked from Blackboard).

To test, input $\phi = 0.07$, $\alpha_1 = \alpha_2' = \alpha_3 = 45°$, $\lambda = 0.95$, and $\eta = 0.92$, leaving $\gamma = 1.4$ and $U/c_0 = 0.4$. Refresh the output. The tool should show $\psi = 0.817$ and $p_{03}/p_{01} = 1.179$.

Make a table showing various possible combinations of the angles and corresponding pressure ratios and de Haller numbers, as obtained from the calculator. Compare your largest pressure ratio result satisfying the de Haller constraints to the result of at least one of the other students (or just post it on Ed!), but without showing the values of the angles. Whose result is better?

The calculator has a link to the formulae. Look at them. Can you use them to get a better result? To get the best possible result?

$$V_{1\theta} = U\phi\tan\alpha_1, \quad V_{2\theta}' = U\phi\tan\alpha_2', \quad V_{3\theta} = U\phi\tan\alpha_3$$

$$\Lambda = \frac{h_2 - h_1}{h_3 - h_1} = \frac{2(h_{02} - h_{01}) - (V_2^2 - V_1^2)}{2(h_{03} - h_{01}) - (V_3^2 - V_1^2)} = \frac{2U\Delta u - (V_{2\theta}^2 - V_{1\theta}^2)}{2U\Delta u - (V_{3\theta}^2 - V_{1\theta}^2)}$$

$$\left.\begin{aligned} \frac{V_2'}{V_1'} &= \frac{\cos\alpha_1'}{\cos\alpha_2'} = \frac{\phi}{\sqrt{\phi^2 + (1 - \phi\tan\alpha_1)^2}}\frac{1}{\cos\alpha_2'} \\ \frac{V_3}{V_2} &= \frac{\cos\alpha_2}{\cos\alpha_3} = \frac{\phi}{\sqrt{\phi^2 + (1 - \phi\tan\alpha_2')^2}}\frac{1}{\cos\alpha_3} \end{aligned}\right\} \geq 0.72 \text{ while } \alpha_1 = \alpha_3$$

$$\frac{c_p(T_{03} - T_{01})}{U^2} = \psi = \lambda\frac{\Delta u}{U} = \overset{\text{fixed}}{\lambda}(1 - (\tan\alpha_1 + \tan\alpha_2')\overset{\text{fixed}}{\phi})$$

$$\frac{p_{03}}{p_{01}} = \left(1 + \eta\frac{T_{03} - T_{01}}{T_{01}}\right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \eta\psi\frac{U^2}{c_p T_{01}}\right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \eta\psi(\gamma - 1)\frac{U^2}{c^2}\right)^{\frac{\gamma}{\gamma-1}}$$

fixed

① $\frac{V_2'}{V_1'}$ and $\frac{V_3}{V_2} \geq 0.72$

② minimise $(\tan\alpha_1 + \tan\alpha_2')$, by MATLAB.

Let $\alpha_1 = \alpha_2'$,

smaller the angle, better the results